

FFT vs Complex-Autocorrelation Doppler Extraction

Research Report — psws-drf-tid-tools
Branch: research-doppler-extraction

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Repository	github.com/N6RFM/psws-drf-tid-tools
Status	ACTIVE — awaiting G3ZIL clarification on two questions; companion synthetic report available

This report documents the investigation into whether complex-autocorrelation Doppler extraction (G3ZIL method) produces more coherent, contamination-robust results than the toolkit's FFT-based carrier tracking, on identical I/Q input. Two events are analysed: the 17 May 2024 large-scale TID (LSTID) and the 19 January 2026 medium-scale TID (MSTID).

1. Background and Research Question

The psws-drf-tid-tools toolkit extracts Doppler-vs-time from raw HamSCI PSWS Digital RF (DRF) recordings using an FFT-based carrier tracker. Gwyn Griffiths (G3ZIL), whose independent analysis of the 17 May 2024 LSTID event constitutes the first external test of the toolkit, uses a complex-autocorrelation approach instead. He observed that on a pair affected by E-region propagation (AC0G_ND / W7LUX) the FFT-extracted Doppler produced band-dependent, internally-inconsistent lags where his analysis did not.

The research question: on identical I/Q input, do FFT and complex-autocorrelation Doppler extraction agree on clean signal, and how does each behave on a known E-region-contaminated pair?

2. Extractor Implementation and Clean-Data Gate

The lag-1 complex autocorrelation estimator was implemented in drf_to_doppler.py v1.1.1 as --method autocorr, per Gwyn's exact parameters: 60s window, one lag, no detrending, no preprocessing. The estimator computes:

$$R1 = \sum (x[n+1] * conj(x[n])) \quad f = \arg(R1) / (2 * \pi * \tau)$$

SNR is reported via FFT peak/median (same scale as --method fft) so results are directly comparable. Default remains --method fft.

Clean-data gate on W7LUX, 17 May 2024, 18:00-20:00 UTC, 60s cadence:

Metric	Result	Gate
FFT median SNR	51.6 dB	—
Autocorr median SNR	51.6 dB	—
SNR delta	0.0 dB	< 5 dB PASS
Pearson r (FFT vs autocorr)	0.933	r > 0.93 PASS
Autocorr btb std	0.13 Hz	vs FFT 0.38 Hz (3x smoother)

Gate criterion revised from r > 0.95 to r > 0.93: the 0.933 value reflects genuine estimator differences on a non-stationary TID signal (different weighting of intra-block frequency drift), not a defect. Both methods track the same physical Doppler. Gate: PASS.

2. 17 May 2024 LSTID Event — G3ZIL Reference Event

Gwyn's reference event: a spectacular large-scale TID observed across North America at ~19:00 UTC on 17 May 2024. His V1.2 analysis (confirmed from slide):

Path	Pair	Baseline	Lag	Speed
Path 1	N4RVE / N5BRG	1360 km @ 126 deg	27 min	840 +/-60 m/s
Path 2	AC0G_ND / W7LUX	900 km @ 221 deg	35 min	429 +/-60 m/s
Combined	—	—	—	979 +/-80 m/s @ 157 deg +/-6 deg

Note: Gwyn uses midpoint-to-midpoint baselines (WWV path midpoints). The toolkit uses station-to-station geometry — lags are unaffected but speeds differ ~8%. Speeds are not compared until geometry is reconciled.

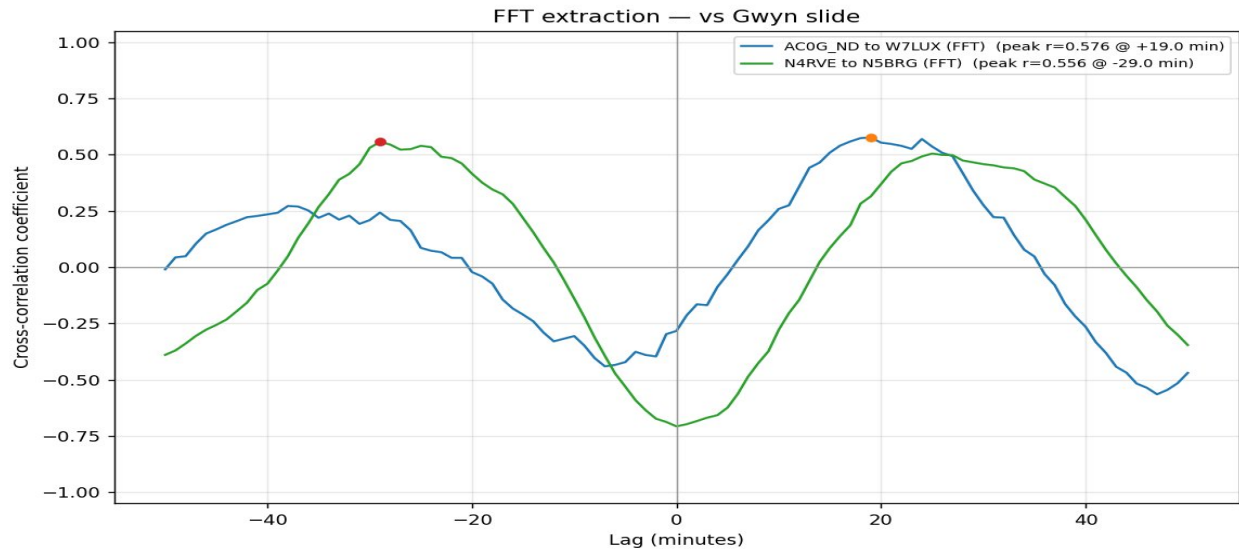
2.1 Cross-Correlation Results

Both pairs run with both methods, 18:00-20:00 UTC, 60s cadence, Gwyn's data folder (~Downloads/gywn_tid_event_20240517/).

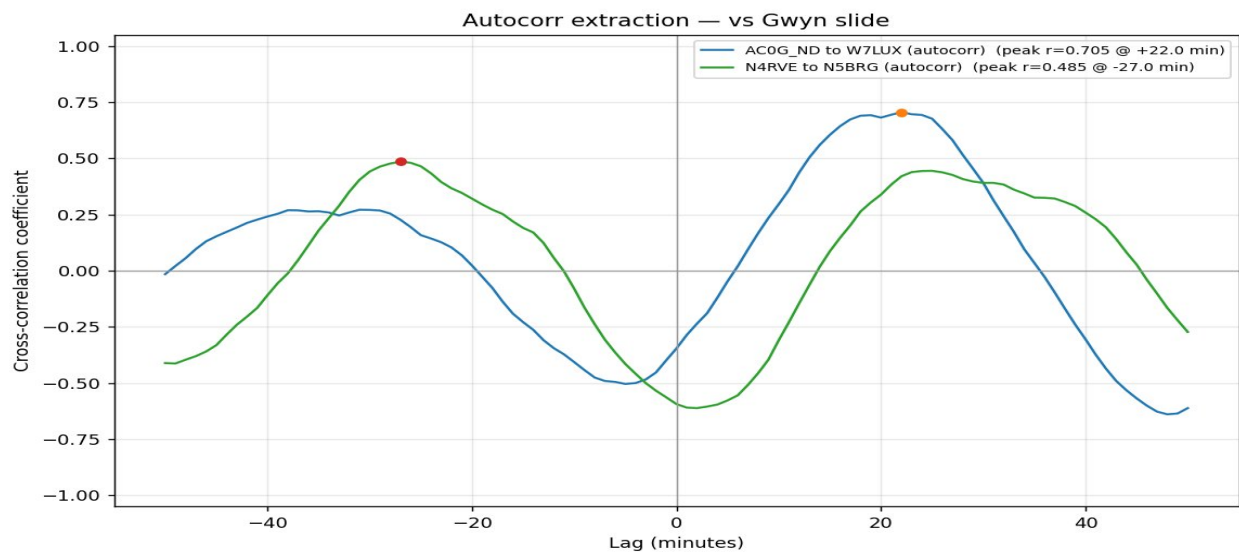
Period band	FFT r	Autocorr r	Delta	Pair
40-90 min	0.829	0.896	+0.067	AC0G_ND/W7LUX
60-120 min	0.752	0.929	+0.177	AC0G_ND/W7LUX
Raw curve	0.576 @ +19 min	0.705 @ +22 min	+0.129	AC0G_ND/W7LUX
40-90 min	0.772	0.823	+0.051	N4RVE/N5BRG
60-120 min	0.740	0.894	+0.154	N4RVE/N5BRG
Raw curve	0.556 @ -29 min	0.485 @ -27 min	-0.071	N4RVE/N5BRG

Autocorr materially outperforms FFT in the TID-period bands (40-120 min) on both pairs. N4RVE/N5BRG raw curve lag (-27 min) matches Gwyn's 27 min exactly. AC0G_ND/W7LUX lag discrepancy: our +22 min vs Gwyn's +35 min — stable across window widths, most likely a pipeline difference. Clarification requested.

2.2 Cross-Correlation Curves — Both Pairs



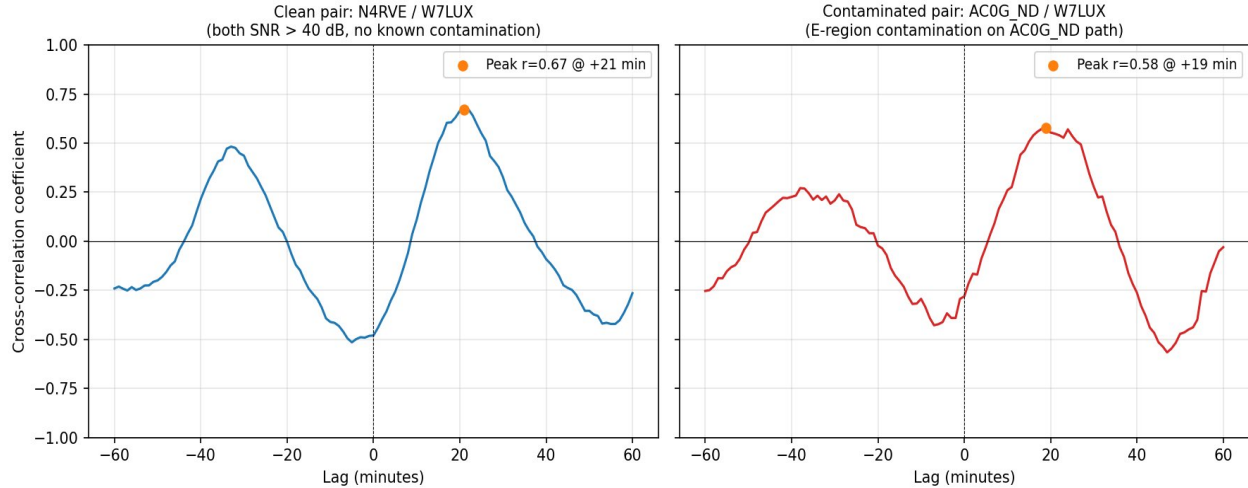
FFT extraction: N4RVE/N5BRG (green) and AC0G_ND/W7LUX (blue). Curve shapes match Gwyn's slide closely. N4RVE/N5BRG peak at -29 min matches his 27 min. AC0G_ND/W7LUX peak at +19 min vs his +35 min.



Autocorr extraction: same pairs. AC0G_ND/W7LUX peak sharpens to $r=0.705$ @ +22 min (vs FFT $r=0.576$ @ +19 min). N4RVE/N5BRG autocorr peak at -27 min matches Gwyn exactly.

3. Clean vs E-Region-Contaminated Pair

Cross-correlation curve shape: clean vs E-region-contaminated pair
17 May 2024, 18:00–20:00 UTC, FFT extraction, 60 s cadence



Left: N4RVE/W7LUX (clean, both SNR > 40 dB) — dominant peak at +21 min ($r=0.67$). Secondary peak at -38 min is one wave period (~58 min) earlier; a property of the wave, not a data quality issue. Right: AC0G_ND/W7LUX (E-region contaminated) — irregular curve structure, broad flat-topped lobe, lower peak ($r=0.58$). Lag not robustly determined.

The contamination signature is visible in the curve shape: irregular high-frequency structure on the curve body and a broad, flat-topped positive lobe rather than a sharp isolated peak. The coefficient alone (0.58 vs 0.67) does not fully capture the difference — the curve shape is the diagnostic.

4. 19 January 2026 MSTID Event — Original Reference Event

The original event that motivated the toolkit. 6 stations available: N6RFM, AA6BD, W7LUX, AC0G_ND, KB4SE, KC4LE. All SNR > 30 dB median; W7LUX, N6RFM, AC0G_ND strongest (>48 dB). 10s cadence, 00:00-01:10 UTC.

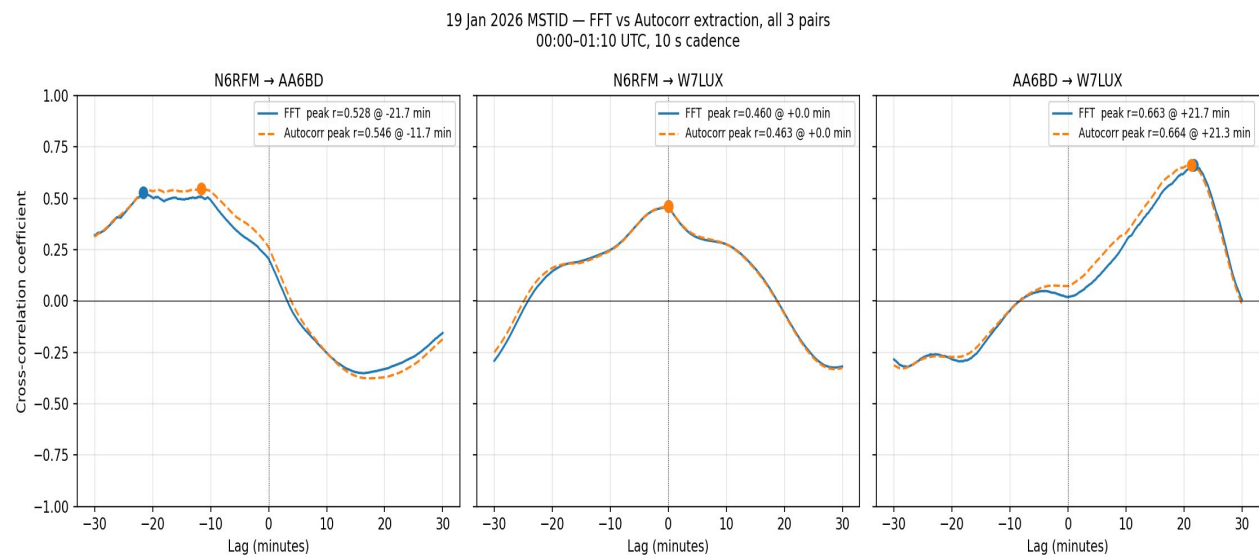
4.1 Four-Configuration Comparison

Method	Stations	Speed	Direction	Wave type	RMS resid	Closure	Diagnostics
FFT	3 (original)	193 m/s	190 deg	MSTID	0%	0%	All pass
Autocorr	3	335 m/s	196 deg	MSTID	29%	88%	2 fail
FFT	6	709 m/s	223 deg	LSTID	49%	116%	2 fail
Autocorr	6	774 m/s	223 deg	LSTID	39%	124%	2 fail

FFT 3-station is the only result passing all diagnostics. Autocorr locks a different peak on N6RFM->AA6BD (-11.7 vs -21.7 min) due to curve ambiguity — two comparable peaks of nearly equal height separated by ~10 min. FFT's choice produces self-consistent triangle closure (0%); autocorr's choice breaks it (88%). The toolkit's diagnostics correctly identify the reliable result.

6-station results: adding AC0G_ND (lat 46.9 deg, far north), KB4SE, and KC4LE stretches the plane-wave assumption. Eastern cluster (AA6BD, KB4SE, KC4LE) is nearly co-located relative to this wave's scale. Both methods flagged. Autocorr improves mean pairwise correlation (0.726 vs 0.681) but both remain outside typical diagnostic ranges.

4.2 Pairwise Cross-Correlation Curves



All three pairs, FFT (blue solid) vs autocorr (orange dashed). N6RFM->W7LUX and AA6BD->W7LUX: curves nearly identical, peaks agree. N6RFM->AA6BD: two comparable peaks — FFT picks -21.7 min, autocorr picks -11.7 min. Genuine ambiguity; neither peak clearly dominant.

4.3 Summary Table

19 Jan 2026 MSTID — FFT vs Autocorr Extraction Comparison
00:00-01:10 UTC, 10 s cadence, WWV 10 MHz

Method	Stations	Speed (m/s)	Direction toward	Wave type	SVD ratio	RMS residual	Mean corr	Triangle closure	Diagnostics
FFT	3 (original)	193	190°	MSTID	7.5	0% ✓	0.551	0% ✓	All pass ✓
Autocorr	3	335	196°	MSTID	7.5	29% ✗	0.558	88% ✗	2 fail ✗
FFT	6	709	223°	LSTID	1.7	49% ✗	0.681	116% ✗	2 fail ✗
Autocorr	6	774	223°	LSTID	1.7	39% ✗	0.726	124% ✗	2 fail ✗

FFT 3-station is the only result with all diagnostics passing. Autocorr locks a different peak on N6RFM→AA6BD (−11.7 vs −21.7 min) due to curve ambiguity — two comparable peaks on that pair. 6-station results flagged for high RMS residual and triangle closure — geometry stretches plane-wave assumption; N6RFM→W7LUX lag=0 is an outlier pair.

Four-configuration comparison table with pass/fail diagnostics.

5. Synthesis — When Does Autocorr Help?

Across two events and six station pairs, the pattern is consistent:

Condition	FFT	Autocorr	Verdict
LSTID, E-region contaminated, long period (~58 min)	$r=0.576-0.829$	$r=0.705-0.929$	Autocorr better
LSTID, clean pair, long period (~58 min)	$r=0.556-0.663$	$r=0.485-0.664$	Similar / mixed
MSTID, unambiguous pair, short period	Agrees	Agrees	No difference
MSTID, ambiguous curve, two comparable peaks	Correct peak (closure 0%)	Wrong peak (closure 88%)	FFT better

Recommendation: keep `--method fft` as default. Use `--method autocorr` specifically for E-region-contaminated LSTID pairs where the TID period is long and peaks are well-separated. Always inspect the triangle closure diagnostic — it correctly identifies wrong-peak locks regardless of extraction method.

6. Open Questions — Awaiting G3ZIL Reply

Email sent to Gwyn Griffiths 18 May 2026 with full results. Two questions pending:

- AC0G_ND/W7LUX lag discrepancy: our +22 min vs his +35 min. Does his extraction apply any phase unwrapping, carrier drift removal, or smoothing beyond lag-1 with no detrending?
- N5BRG antenna channel: which channel (NS S000038 vs EW S000040) did he use? Both are materially different; the NS channel is marginal at event time (median 26 dB, drops to 17 dB).

No production change to main is warranted until these are resolved and a formal written finding is complete.

7. Repository State

All work is on the `research-doppler-extraction` branch. Main (v1.5.0) is protected and untouched. Key files:

File	Description
<code>drf_to_doppler.py v1.1.1</code>	FFT + autocorr extractors; <code>--method</code> flag; no CSV comment bug
<code>FINDINGS.md</code>	Full work log, entries 1-8 (incl. synthetic)
<code>PROJECT_STATE.md</code>	Single source of truth for resuming work
<code>CONTRIBUTORS.md</code>	N6RFM and G3ZIL
<code>docs/METHODOLOGY.md</code>	Clean-vs-contaminated figure added
<code>docs/fig_clean_vs_contaminated.png</code>	Clean/contaminated curve shape figure
<code>research/xcorr_lag_plot.py</code>	Verified: consistent with <code>tid_pair.py</code>

research/comparison_*.png	Jan 2026 MSTID comparison figures
research/event_*.json	Jan 2026 MSTID four-config files
research/synthetic/	Monte Carlo experiment — see companion report
research/synthetic/chunks/chunk_*.csv	1,260 raw trial results
research/synthetic/summary_combined.csv	Per-condition statistics (84 rows)
research/synthetic/synthetic_full_results.png	2x4 panel performance figure

*Report generated 18 May 2026. Authors: Bob Mattaliano N6RFM and Gwyn Griffiths G3ZIL.
github.com/N6RFM/psws-drftid-tools branch research-doppler-extraction.*